

IRKUTSK, Russian Federation

WMO: 307100

Lat: 52.2721N

Lon: 104.3080E

Elev: 1539

StdP: 13.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-25.8	-38.1	0.7	-31.8	-31.4	1.0	-25.4	21.9	6.1	19.6	6.3	3.9	90	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	84.4	64.2	80.9	63.5	78.3	62.4	68.0	78.5	66.3	76.0	64.7	74.0	7.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	96.5	71.5	62.8	90.5	70.1	61.1	85.0	68.7	33.3	78.6	31.9	75.9	30.6	73.3	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.0	16.4	-35.9	90.2	7.2	2.6	-41.1	92.1	-45.3	93.6	-49.3	95.0	-54.6	96.9
			-36.1	71.6	7.2	2.3	-41.2	73.3	-45.4	74.6	-49.4	75.8	-54.6	77.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	-2.3	4.4	19.9	37.4	49.3	60.8	65.2	60.7	48.6	34.3	15.1	2.3
	DBStd	25.28	12.09	11.36	11.35	8.10	7.63	6.61	4.94	5.68	7.06	8.88	12.85	12.22
	HDD50	7455	1620	1277	934	383	109	5	0	2	110	489	1048	1478
	HDD65	11752	2085	1697	1399	827	487	158	59	154	493	952	1498	1943
	CDD50	1296	0	0	0	6	88	328	470	335	67	2	0	0
	CDD65	117	0	0	0	0	1	31	64	21	0	0	0	0
	CDH74	1705	0	0	0	6	123	534	705	309	28	0	0	0
	CDH80	392	0	0	0	0	26	141	169	53	3	0	0	0
Wind	WSAvg	6.7	5.3	5.8	6.8	8.5	8.6	7.3	6.2	6.5	6.8	6.8	6.2	5.1
Precipitation	PrecAvg	18.5	0.5	0.4	0.5	0.7	1.3	2.5	4.4	3.5	2.0	1.0	0.7	0.8
	PrecMax	24.9	1.2	0.9	1.6	2.5	2.6	5.7	13.9	8.7	5.1	3.5	1.5	2.2
	PrecMin	12.6	0.1	0.0	0.0	0.3	0.3	0.3	0.9	0.9	0.5	0.2	0.3	0.1
	PrecStd	3.3	0.3	0.2	0.3	0.5	0.7	1.3	2.2	1.9	1.0	0.6	0.3	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.0	37.2	53.5	71.7	83.8	89.2	88.3	84.7	77.4	64.8	46.4	31.6
		MCWB	24.5	29.6	40.0	49.2	57.2	64.6	67.2	66.0	60.1	49.3	38.5	28.1
	2%	DB	22.7	29.6	45.8	64.3	77.1	84.1	84.5	80.8	71.9	57.5	40.0	26.5
		MCWB	20.8	24.8	35.9	46.0	54.7	62.7	66.0	64.9	57.6	45.5	34.5	24.4
	5%	DB	18.0	24.8	41.0	58.4	71.9	80.4	80.9	77.2	67.7	53.0	35.8	22.8
		MCWB	16.4	21.4	33.3	43.5	52.2	61.7	65.3	63.4	55.3	43.2	32.2	21.4
	10%	DB	14.1	20.8	37.1	53.2	67.1	76.6	77.9	73.7	62.8	48.2	32.2	19.4
		MCWB	13.0	18.0	30.8	40.8	49.8	60.5	64.4	62.4	52.8	41.0	29.9	18.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	29.5	40.5	51.1	59.4	68.2	71.4	68.9	62.4	50.1	39.1	28.2
		MCDB	26.8	36.3	52.4	69.4	80.0	81.3	82.0	79.5	73.2	62.6	45.6	30.7
	2%	WB	20.5	24.9	36.4	46.8	55.9	65.7	69.0	66.6	59.3	46.5	35.1	24.3
		MCDB	22.3	29.0	44.7	62.0	73.4	77.9	79.5	76.9	68.7	56.0	39.3	26.0
	5%	WB	16.6	21.2	33.5	43.9	53.4	63.7	67.4	64.9	56.5	44.0	32.4	21.3
		MCDB	17.9	24.5	39.9	57.2	68.7	75.5	76.9	74.6	65.8	51.4	35.2	22.7
	10%	WB	12.9	17.8	30.8	41.1	50.8	61.8	65.7	63.2	53.8	41.4	30.0	18.3
		MCDB	13.9	20.5	36.2	52.0	64.5	73.3	74.4	71.8	61.2	47.7	32.4	19.4
Mean Daily Temperature Range	5% DB	MDBR	18.0	22.9	23.8	22.5	26.1	24.4	21.6	21.5	22.2	20.2	18.8	17.1
		MCDBR	19.3	25.3	26.2	33.7	36.8	33.0	29.4	29.2	31.4	29.1	20.5	18.2
		MCWBR	17.5	20.9	17.5	18.0	18.1	14.8	13.7	14.9	17.3	17.7	15.5	16.5
	5% WB	MCDBR	19.0	25.1	24.3	30.9	32.5	26.9	23.6	25.1	27.9	27.0	19.3	17.7
		MCWBR	17.3	20.8	16.3	17.3	16.6	12.9	12.2	13.4	15.8	17.0	15.0	16.3
Clear-Sky Solar Irradiance	taub	0.276	0.302	0.338	0.385	0.445	0.439	0.429	0.395	0.342	0.322	0.299	0.279	
	taud	2.085	2.068	2.111	2.214	2.064	2.193	2.266	2.343	2.445	2.395	2.201	2.071	
	Ebn at Noon	220	251	266	268	256	259	260	263	267	247	211	193	
	Edh at Noon	24	33	39	40	49	44	40	35	28	24	21	20	
All-Sky Solar Radiation	RadAvg	404	756	1219	1538	1757	1856	1715	1463	1113	707	403	283	
	RadStd	21	30	53	87	96	102	129	73	103	40	25	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page